Analysis 1: Aug-Nov 2026

Prove the assertions from first principles.
You can use results on convergence of sequences; continuity and uniform continuity of functions (but give precise statement)
Do not use any other standard result(s) without providing proof.

Exercises

(I) Let $f(x) = x^3 - 5x^2 + 28$ and $g(x) = x^4 + 22x + 1$ for $x \in [0, \infty)$. Let $p(x) = g(f(x))$. Let $\epsilon = 0.001$. Let $x^* = 7$. Get $\delta > 0$ such that

$$0 < |h| < \delta \text{ implies } \left| \frac{p(x^* + h) - p(x^*)}{h} - p'(x^*) \right| < \epsilon.$$

Here p' denotes the derivative of p. You have to produce a value of δ that works.

(II) For $n \geq 1$ let f_n : be a real valued continuous function on $[-1, 1]$ and differentiable on $(-1, 1)$, with derivative f_n'. Suppose f_n' is continuous on $(-1, 1)$ for all n. Suppose $\sum_n f_n(x)$ converges uniformly to $f(x)$ on $[-1, 1]$ and $\sum_n f_n'(x)$ converges uniformly to $g(x)$ on $(-1, 1)$. Show that f is differentiable on $(-1, 1)$, with derivative $f' = g$.

(III) Let $< a_n : n \geq 1 >$ be a bounded real valued sequence. Show that :
[a] For all $x \in \mathbb{R}$, show that the series $\sum_n a_n \frac{x^n}{n!}$ converges.
[b] Let $f(x) = \sum_n a_n \frac{x^n}{n!}$. Show that f is a continuous function on $\mathbb{R}$.
[c] f defined above is differentiable for all $x \in \mathbb{R}$ and obtain the power series of
$g = f'$, which converges for for all $x \in \mathbb{R}$.
[d] g defined above is differentiable and obtain the power series of $h = g'$.

(IV) For $x \in \mathbb{R}$, let $e(x) = 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!}$. Show that
[a] $e(\cdot)$ is well defined $\forall x \in \mathbb{R}$ and $e(0) = 1$.
[b] $\forall\, x, y \in \mathbb{R}$, we have $e(x + y) = e(x)e(y)$.
[c] $e(x) > 0$ for all $x \in \mathbb{R}$.
[d] If $f : \mathbb{R} \mapsto \mathbb{R}$ is a differentiable function such that $f(0) = 1$ and $f'(x) = f(x)$ for all x, then $f(x) = e(x)$ (*In other words, the differential equation $f' = f$ with boundary condition $f(0) = 1$ admits a unique solution, namely $f = e$*).